Photon Shot Noise Limits on Optical Detection of Neuronal Spikes and Estimation of Spike Timing

Paper: Wilt, Fitzgerald and Schnitzer (2013), *Biophysical Journal* 104:51–62. Theory, simulations and comparison with published optical recordings.

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Page references below use PDF page numbers.

To which general question does this paper relate?

How accurately can optical recordings reveal neuronal spikes, and what does this imply for estimating population firing-rate states? It concerns measurement, not stimulation or synthesis of a neural controller.

How does it answer the general questions?

Measured signal. Calcium/voltage indicators produce fluorescence from which spikes are inferred. Photon counts are Poisson-distributed: their variance equals their mean. The single-spike model is $F(t) = F_0 + Ae^{-t/\tau}\mathbf{1}_{t\geq 0}$, where F_0 includes baseline and unwanted background, A is the fluorescence increment and τ its decay time (PDF pp. 1–2).

Usable insight. For small relative transients, sufficient photon counts and sampling rate ν with $\nu\tau > 1$, spike discriminability is

$$d' \approx \frac{\Delta F}{F}\sqrt{\frac{F_0\tau}{2}} = A\sqrt{\frac{\tau}{2F_0}}.$$

Larger relative responses and less unwanted background improve detection. Longer transients collect more photons but impair timing and separation of nearby spikes. Faster sampling alone does not ensure better estimates. Likelihood-based detection quantifies missed-spike/false-alarm tradeoffs (PDF pp. 3–7, 9–11).

Timing and feedback. Detection and timing are distinct tasks. For $d' = 5$, $\tau = 0.15$ s and $\nu = 20$ Hz, the calculated timing-error standard-deviation lower bound is 20.7 ms. This bound applies to unbiased estimators and need not be attained; it is not an online processing-delay guarantee (PDF pp. 6–7, 10).

Model relevance (derived). A possible population-rate observation approximation, with r the firing-rate state and c an indicator state, is

$$\dot{c} = -c/\tau + qr, \qquad Y_k \sim \text{Poisson}\left(\int_{t_k}^{t_{k+1}} [F_0 + \alpha c(t)]\, dt\right).$$

This motivates an observer that includes indicator dynamics and counting noise; it does not establish observability. Averaging reconstructed spikes from representative neurons can estimate a population rate. The paper does not validate this population model or direct full-state feedback.

Limits. The main analysis concerns isolated/sparse spikes, linear unsaturated responses, known waveforms and already separated cells. Motion, optical mixing, saturation and overlapping spikes can worsen performance. The bounds are ideal limits; technology comparisons describe 2013 devices, rather than current optical capability (PDF pp. 2, 7, 9–11).

What papers does it cite that could provide more information?

- Yaksi and Friedrich (2006), *Reconstruction of firing rate changes across neuronal populations by temporally deconvolved Ca^{2+} imaging* (ref. 24): directly relevant to population-rate reconstruction.
- Vogelstein et al. (2010), *Fast nonnegative deconvolution for spike train inference from population calcium imaging* (ref. 18).
- Grewe et al. (2010), *High-speed in vivo calcium imaging reveals neuronal network activity with near-millisecond precision* (ref. 15).